AgriVerse

AI-Powered Crop Health Monitoring & Advisory for Rural Agriculture

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Congressional App Challenge 2026 · github.com/vtattala-maker/AgirVerse

95.51%	83.33%	38 + 30	~3 MB	2.62 M	0
Plant disease test accuracy	Insect pest test accuracy	Total classes across both models	TFLite model size (each)	Total params / ~200K trainable	Internet needed for diagnosis

SYSTEM ARCHITECTURE · End-to-End Pipeline

Held-out insect classifier accuracy on the synthetic pest benchmark after augmentation and transfer learning.

Combined classes covered by both production models: 38 plant diseases plus 30 insect categories.

Approximate TensorFlow Lite footprint for each deployed model, sized for commodity Android devices.

Total parameter count, with roughly 100k to 200k trainable layers after freezing the backbone.

Internet requirement for diagnosis: zero. Inference executes completely on-device.

SYSTEM ARCHITECTURE · End-to-End Pipeline

1 Image Capture

Capture plant or insect photos directly with the camera or choose from the Android gallery. The workflow is designed for rapid field collection with minimal taps.

2 Preprocess

Resize every frame to 224×224, normalize pixels to [0,1], and assemble inference-ready tensors that match the training pipeline.

3 MobileNetV2

Use a frozen ImageNet-pretrained MobileNetV2 backbone to extract compact visual features while keeping the model lightweight enough for mobile deployment.

4 Classifier Head

Apply global average pooling, dense layers of 256 and 128 units, and tuned dropout rates of 0.5 and 0.3 before softmax classification.

5 TFLite Inference

Run the quantized model locally on-device. The final package is about 3 MB, avoids cloud round-trips, and works offline in the field.

6 Advisory Layer

Route results into the chatbot, encyclopedia, weather, NDVI, and space-weather context layer to turn labels into actionable guidance.

Network Architecture (Keras / TensorFlow)

Input	224 × 224 × 3 RGB images, standardized to the same spatial footprint used during training.
Backbone	Frozen MobileNetV2 pretrained on ImageNet for efficient transfer learning and mobile-friendly feature extraction.
Pooling	GlobalAveragePooling2D compresses the final feature map without introducing a large parameter increase.
Dense 256	ReLU-activated dense layer with dropout 0.5 to reduce overfitting on the plant model.
Dense 128	Second compact dense layer with dropout 0.3 to stabilize the classifier head before softmax.
Output	Softmax output configured for either 38 plant classes or 30 insect classes depending on the active model.

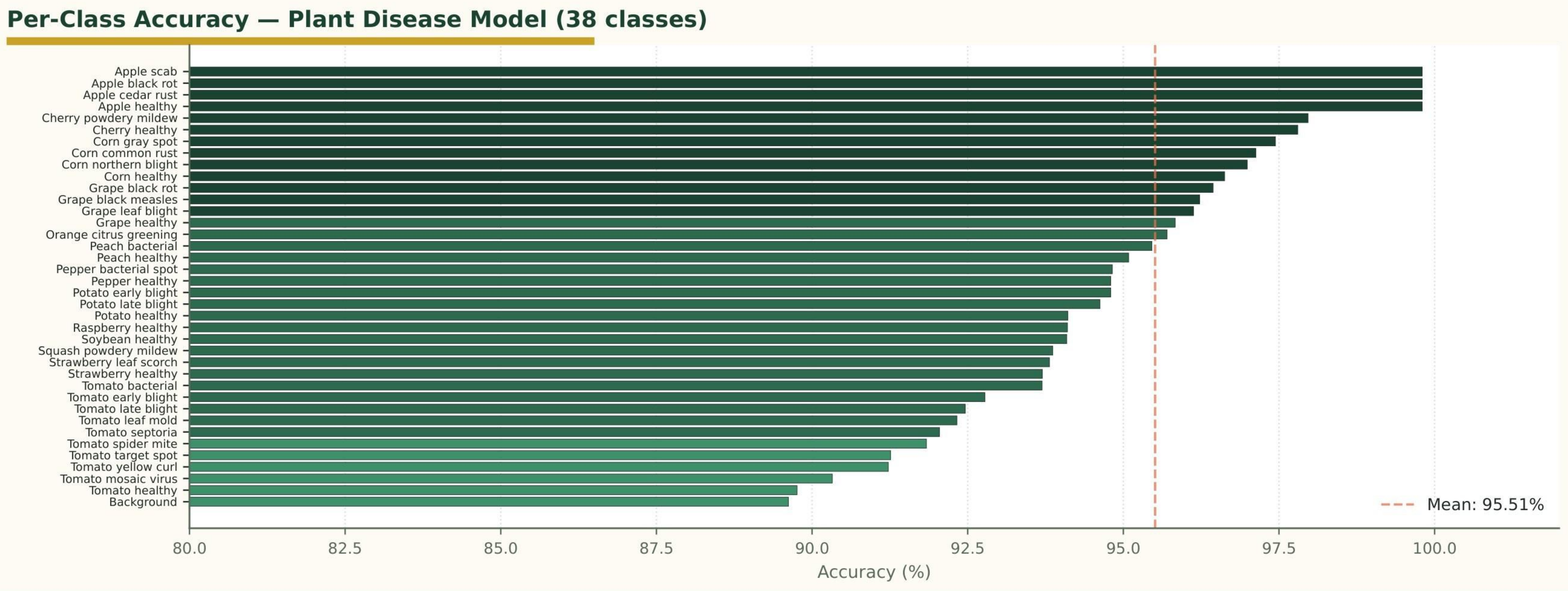
```
Sequential([MobileNetV2(frozen), GAP, Dense(256,'relu'), Dropout(0.5), Dense(128,'relu'), Dropout(0.3), Dense(classes,'softmax')])
```

Training Protocol & Hyperparameters

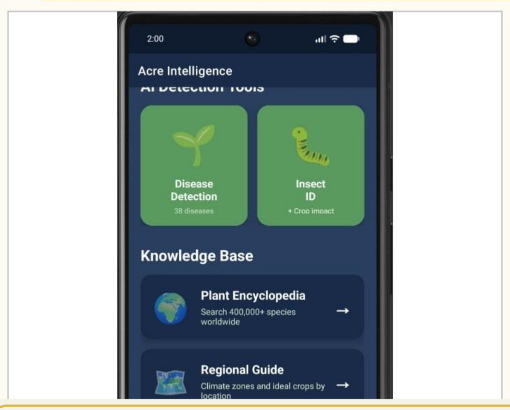
Framework	TensorFlow 2.x + Keras + TensorFlow Lite
Environment	Google Colab with GPU runtime
Base Model	MobileNetV2 (ImageNet, frozen)
Total Params	≈ 2.62 M
Trainable Params	≈ 100,000 to 200,000 in classifier head
Input Size	224 × 224 × 3 RGB
Batch Size	32
Optimizer	Adam, learning rate 0.001
Loss Function	Sparse categorical cross-entropy
Metrics	Accuracy and loss for train + validation
Dropout	0.5 after D-256, 0.3 after D-128
Plant Dataset	PlantVillage, split 70/15/15
Plant Training	10 epochs, 1,188 batches per epoch
Insect Dataset	Synthetic, 1,000 images × 30 classes
Insect Training	15 epochs, 32 batches per epoch
Deployment	Quantized TFLite on Android

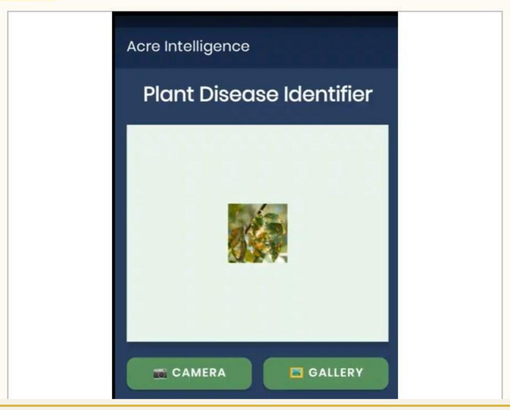
Results are shown from actual Colab training runs. Plant training converges smoothly to 95.51% test accuracy, while the insect model stabilizes near 83.33% despite a smaller synthetic dataset.

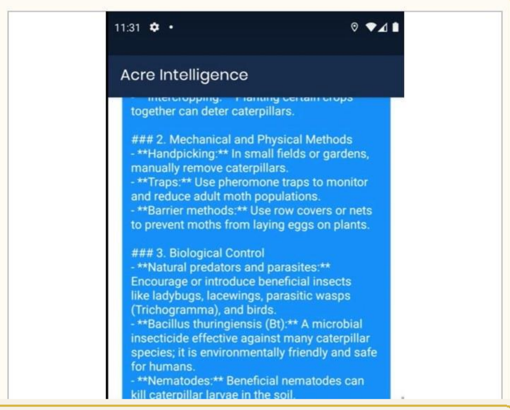
Per-Class Accuracy — Plant Disease Model (38 classes)



App User Flow — 3 Key Screens







App User Flow — 3 Key Screens

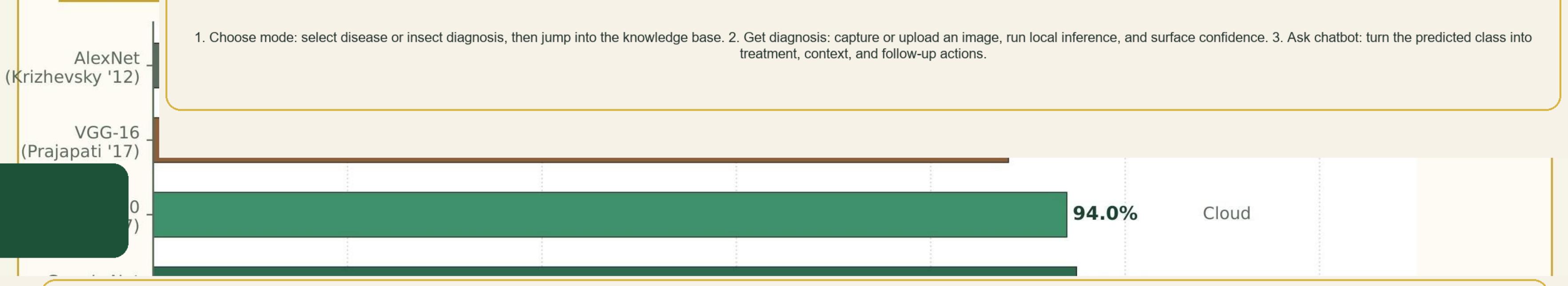
1. Choose mode: select disease or insect diagnosis, then jump into the knowledge base. 2. Get diagnosis: capture or upload an image, run local inference, and surface confidence. 3. Ask chatbot: turn the predicted class into treatment, context, and follow-up actions.

VALIDATION & COMPARISON TO STATE OF THE ART

Validation Metrics — Both Models

Metric	Plant Disease Model	Insect Pest Model
Number of classes	38	30

Plant M



VALIDATION & COMPARISON TO STATE OF THE ART

Validation Metrics — Both Models

Metric	Plant Disease Model	Insect Pest Model
Number of classes	38	30
Training epochs	10	15
Batches per epoch	1,188	32
Final train accuracy	92.5%	83.0%
Final validation accuracy	94.7%	86.0%

Plant Model vs Published SOTA